Class Meeting Handout #4

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Understanding mechanism can help us understand the outcomes of reactions and propose schemes that explain observations. Then we will need to test these hypotheses.

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Selected Mechanisms in Synthesis

After recalling some basic building blocks of mechanism in the previous meeting, let is go a little farther and interpret some observations.[1] The reactions below are examples of steps in syntheses of larger, more complex targets.

[1] Hoffman's *Organic Chemistry: an Intermediate Text* would be a good place to look for review material. Chapters 7 & 8 are recommended.

Review: Mechanism Practice – 40 minutes

1. Propose a synthesis for making an alkyl bromide form an alkane. Propose a synthesis for making an alkene from an alkyl bromide. Propose a synthesis for making an epoxide from an alkene.[2]

[2] There was a time when all you chemicals had to be made from petroleum. Imagine spending the first two years of your Ph.D. making basic starting materials that today you can have shipped to you overnight.

2. When using isotopically labeled water, the hydrolysis of epoxide gives different products depending on if the reaction is performed in acidic or basic solution (see figure 1). Can you explain the observation by proposing a mechanism for each reaction? Predict the stereochemistry in each case. How would you exploit this observation in organic synthesis?

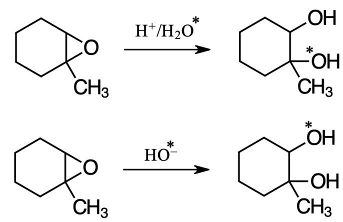

Figure 1: Acid and base catalyzed hydrolysis of an epoxide

3. Explain the mechanisms that give rise to alcohol and ether products below. What reaction conditions would you choose to maximize the yield of each?

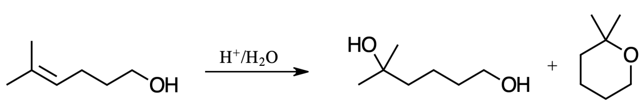

Figure 2: Acid catalyzed addition to an alkene

4. Bromination of acetone in water is fast but the rate is independent of bromine concentration. (See figure 3). Can you explain the observation by proposing a mechanism for the reaction?

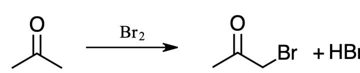

Figure 3: α–bromination of acetone

5. The benzoin condensation seems to be impossible. And it is unless catalyzed by cyanide ion. Propose a mechanism. Be able to argue for the stability of your intermediates.

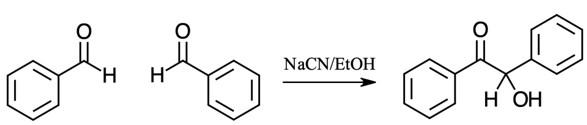

Figure 4: The benzoin condensation

6. Make a carbon–carbon bond where the product contains an alcohol group at the new connection using a mixed aldol addition reaction.

7. Make a carbon–carbon bond where the product contains a ketone group at the new connection using a mixed Claisen condensation reaction.[3]

8. Make a carbon–carbon bond where the product contains only alkane carbons at the new connection. Use any method you like, but some are much better than others.

[3] Why is "Claisen" capitalized while "aldol" is not?

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing the requirements described on the moodle site you will have achieved the following...

1. Have begun to recall the chemistry that you learned in the first three years of your time at UPEI.

2. Be able to propose reaction mechanisms when given starting material and product.

Looking Ahead

We will be using the skills that we reviewed today and in the previous meeting throughout the rest of this course. Being able to correctly propose a mechanism will enable you to better test your hypotheses as you investigate mechanisms for fun and profit.

In the next meeting we will complete our review of structure and reactivity by using what we have recalled about conformation, molecular orbitals and mechanism to review what you have learned about transition state structure. The assignment, due before the next meeting, will be important in this final review. We will also introduce a new concept, a 3-dimensional reaction coordinate diagram called a More O'Ferrall-Jenck's plot.